Method of Transformations (Bivariate)

Consider $U = h_1(X,Y)$ and $V = h_2(X,Y)$ to be two functions of the continuous random variables X and Y, with joint density $f_{X,Y}(x,y)$.

If the transformation from (x,y) to (u,v) is one-to-one for all $(x,\ y)$ (where $f_{X,Y}(x,y) > 0$) then the joint density of U and V is

$$f_{U,V}(u,v) = f_{X,Y}\left(h_1^{-1}(u,v), h_2^{-1}(u,v)\right)|J|$$

where

$$J = \det\begin{pmatrix} \dfrac{\partial h_1^{-1}}{\partial u} & \dfrac{\partial h_1^{-1}}{\partial v} \\ \dfrac{\partial h_2^{-1}}{\partial u} & \dfrac{\partial h_2^{-1}}{\partial v} \end{pmatrix} = \frac{\partial h_1^{-1}}{\partial u}\frac{\partial h_2^{-1}}{\partial v} - \frac{\partial h_1^{-1}}{\partial v}\frac{\partial h_2^{-1}}{\partial u} \quad \text{(called the Jacobian)}$$

- The inverse functions are $x = h_1^{-1}(u,v)$ and $y = h_2^{-1}(u,v)$.

Two R.v. x, y

$\Downarrow$

Two R.v. u, v.

$u = h_1(x, y) \qquad v = h_2(x, y)$

<u>Step 1</u> : $x = h^{-1}(u, v) \qquad y = h_2^{-1}(u, v)$

<u>Step 2</u> : Compute the Jacobian Matrix :

$$\begin{bmatrix} \dfrac{\partial h_1^{-1}(u,v)}{\partial u}, & \dfrac{\partial h^{-1}(u,v)}{\partial v} \\ \dfrac{\partial h_2^{-1}(u,v)}{\partial u}, & \dfrac{\partial h_2^{-1}(u,v)}{\partial v} \end{bmatrix}$$

absolute value

<u>Step 3</u> : compute the determinant.

$$A = \left| \frac{\partial h^{-1}(u,v)}{\partial u} \cdot \frac{\partial h_2^{-1}(u,v)}{\partial v} - \frac{\partial h^{-1}(u,v)}{\partial v} \cdot \frac{\partial h_2^{-1}(u,v)}{\partial u} \right|$$

<u>Step 4</u> : density function :

$$f_{u,v}(u,v) = f_{xy}\left(h^{-1}(u,v), h_2^{-1}(u,v)\right) \cdot A$$

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Comments:

1. Once $f_{U,V}(u,v)$ is found, we can find the marginal densities:
   - $f_U(u) = \int_{\text{over } v} f_{U,V}(u,v)dv$
   - $f_V(v) = \int_{\text{over } u} f_{U,V}(u,v)du$

2. If you only have one function of interest, say $U = h_1(X,Y)$, you can create any arbitrary (second) function $V = h_2(X,Y)$ and use the method described on the previous page.

<u>Example 1:</u> (suppose $\theta = 1$)

The joint density of X and Y is given as

$$f(x,y) = \begin{cases} 1\ , & 0 < y < 1 \text{ and } 0 < x < 1, \\ 0\ , & else \end{cases}$$

c) What is the probability distribution of $U = X / Y$?

h) What is the joint probability distribution of $U = X + Y$ and $V = X - Y$?

Given $f_{U,V}(u,\ v)$, find $f_U(u)$ and $f_V(v)$.

Example 6:

Let X and Y be independent and identical $Normal$ random variables with mean μ and variance σ^2.

e) Consider $U = X + Y$ and $V = X - Y$. Find the joint distribution of U and V.

Given $f_{U,V}(u, v)$, find $f_U(u)$ and $f_V(v)$.